Chiral Phase-Momentum Emission (CPME)

Irfan Mahir — Independent Researcher, Dhaka, Bangladesh — irfan@furylogic.com

Contents

Chiral Phase-Momentum Emission (CPME) **1**
0. Position . . . 1
1. Architecture . . . 1
1.1 Chiral Wave Injection (from HTR §2.1) . . . 2
1.2 Momentum Filter (from MCPE §2.1) . . . 2
1.3 Acoustic Mass (from MCPE §2.2) . . . 2
1.4 Settle Separation (from MCPE §2.3) . . . 2
1.5 Coherence-Rise Metric (from HTR §2.2) . . . 3
1.6 Phase-Collapse Projection (from HTR §2.3) . . . 3
2. Composition . . . 3
3. Emission Contract . . . 4
4. Parameters . . . 4
5. Expected Dynamics . . . 4
6. Failure Modes . . . 5
7. Biological Analog . . . 5
8. Contract Summary . . . 6
9. Summary . . . 6

Chiral Phase-Momentum Emission (CPME)

Irfan Mahir *Independent Researcher, Dhaka, Bangladesh* irfan@furylogic.com

A unified emission substrate combining the Momentum-Coupled Phase Emission (MCPE) filter architecture with the Holographic Topological Resonance (HTR) phase-projection readout. CPME operates on a single phase manifold: chirality on injection, coherence-rise as the resonance metric, momentum as the eligibility filter, and continuous phase-collapse sweep as the emission mechanism. No syntax trees, no grammar rules, no next-token statistics, and no fixed output length.

---

0. Position

The Continuous Morphological Dynamics (CMD) substrate, extended by Voltage-Gated Harmonic Topology (VGHT), produces domain-orthogonal cognition with sparse cross-domain analogical bridging. Its emission layer, however, has two orthogonal deficits that the previous literature treats separately:

1. **Stimulus Reflection.** External input directly drives concept cavities to maximal amplitude; the fixed-slot Kuramoto readout locks onto those cavities before internal propagation can produce a downstream response. The system parrots.

2. **Fixed-Slot Rigidity.** The Subject/Verb/Object slot readout presupposes a three-word template. Simpler answers waste slots; more complex answers are truncated. Word order is imposed by the slot definitions rather than emerging from the field.

MCPE addresses (1) by shifting the readout observable from amplitude A to momentum dA/dt and by muting recently-driven cavities via acoustic mass. HTR addresses (2) by replacing the fixed slots with a continuous carrier-phase sweep over a per-cavity phase field, and by introducing arrival-time chirality on injection so that word order becomes a physical gradient rather than a syntax rule.

CPME unifies the two proposals into a single emission substrate. MCPE identifies *which* cavities are eligible to emit; HTR identifies *in what order* eligible cavities emit. The two operate on orthogonal state variables and compose without collision.

---

1. Architecture

Let the substrate be the CMD + VGHT engine as previously specified: cavities indexed by $i \in [0, N)$ with per-cavity displacement x_i, velocity v_i, heat T_i, and quiet age q_i; tunnels with per-tunnel stiffness k, damping c, polarity p, and voltage gating $g \in \{0, 1\}$; and macro-nodes with per-cluster phase θ_m, amplitude A_m, and natural frequency ω_m drawn from the VGHT sub-band ladder. The base cavity frequency is ω_0.

CPME adds six mechanisms.

1.1 Chiral Wave Injection (from HTR §2.1)

Each cavity records the simulation time at which it was most recently driven:

```
t_arrival_i ← sim_time    (whenever drive(i, ·) is called)
```

The cavity's oscillator phase at any subsequent tick is $\phi_i^{\text{driven}} = \omega_0 \cdot (T_{\text{current}} - t_{\text{arrival},i})$. Earlier-arriving bytes therefore accumulate *more* phase; the subject leads in phase because it entered the substrate first. Chirality is inherited *for free* from the existing sinusoidal drive term — the arrival time only needs to be *recorded* so the readout can reference it. No new tick math.

1.2 Momentum Filter (from MCPE §2.1)

Every cavity carries an exponential-moving-average amplitude envelope:

```
E_i ← β · E_i + (1 − β) · |x_i|          (β = 0.9)
```

and the tick publishes the *previous-tick* envelope as E_i^{prev} so that the momentum

```
P_i = E_i − E_i^{\text{prev}}
```

is the low-pass-filtered rate of change of amplitude. Directly-driven cavities that have peaked and are dispersing energy have $P_i \leq 0$; deep-graph cavities absorbing energy from downstream propagation have $P_i > 0$. The emission eligibility factor is $\text{ReLU}(P_i)$, a smooth non-negative scalar.

This replaces the naïve $dA/dt = |v|$ - style derivative, which oscillates at $2\omega_{\text{drive}}$ and produces a strobe artefact in the readout. Envelope EMA is the physically correct low-pass.

1.3 Acoustic Mass (from MCPE §2.2)

Each cavity carries a per-cavity mass increment M_i driven by the external injection load:

```
dM_i/dt = γ · I_{ext, i} − δ · M_i
```

with $\gamma = 0.30$, $\delta = 0.08$. $I_{\text{ext},i}$ is the instantaneous drive envelope on cavity i, i.e., precisely the signal that makes an input-driven cavity "loud." The resulting emission mask

```
μ_i = 1 / (1 + M_i / M_{scale})          (M_scale = 1.0)
```

falls toward zero for freshly-driven cavities and returns toward $\mu_i = 1$ as M_i decays. Applied only in the emission scoring; the field dynamics are untouched, so graph connectivity is preserved.

1.4 Settle Separation (from MCPE §2.3)

The tick maintains a global injection load

```
L = Σ_i drive_amp_i · drive_envelope_i
```

and derives a *slot coupling* factor

```
K_settle = 1 − σ((L − L_thresh) / L_scale)
```

with σ the logistic, $L_{\text{thresh}} = 0.5$, $L_{\text{scale}} = 0.2$. During active injection, $K_{\text{settle}} \approx 0$ and the emission readout is suppressed. As the drive envelopes decay, $K_{\text{settle}} \to 1$ and articulation engages. The transition is continuous — no discrete $T_0/T_1/T_2$ boundary.

1.5 Coherence-Rise Metric (from HTR §2.2)

The tick computes the *localized* Kuramoto order parameter over input-touched macros:

```
R_local = |Σ_{m ∈ touched} A_m · e^{i θ_m}| / (Σ_{m ∈ touched} A_m + ε)
```

where M_{touched} is the set of macros with at least one non-quiet cavity ($q_i < K_{\text{idle}}$). R_{local} is tracked against its EMA baseline $\bar{R}$:

```
ΔR = R_local − R̄          (R̄ ← λR̄ + (1−λ) R_local, λ = 0.98)
```

ΔR is exposed as a substrate observable but does not directly gate emission; it is diagnostic. Silence when ΔR is low emerges from the emission floor being unmet, not from an explicit ΔR gate. This preserves the substrate's honest-silence property without turning ΔR into a rule.

1.6 Phase-Collapse Projection (from HTR §2.3)

The emission readout is replaced entirely. Per-cavity phase is computed on demand from existing state:

```
φ_i = atan2(v_i, ω_0 · x_i)
```

The carrier phase ϕ_{carrier} is swept monotonically across $[0, 2\pi)$ in K_{steps} discrete increments (default $K_{\text{steps}} = 12$). At each swept instant, the cavity chosen for emission is the arg-max of

```
w_i(φ_carrier) = A_i · ReLU(P_i) · μ_i · K_settle
                · exp(−κ_sweep · (φ_carrier − φ_i)^2)
```

where $A_i = |x_i|$ is the instantaneous amplitude and $\kappa_{\text{sweep}} = 4$ sets the Gaussian window width. Winners below the floor $\theta_{\text{emit}} \cdot \max_j w_j$ are silenced (no emission at that sweep step). Winners are suppressed from all subsequent steps of the sweep (each cavity emits at most once per turn). The sweep terminates non-cyclically at $\phi_{\text{carrier}} = 2\pi$, so wrap-around is not a source of duplicate emissions.

Output length is emergent: as many tokens as there are distinct high-weight peaks under the sweep. A sparse field emits few tokens (possibly zero — honest silence); a dense, structurally-resonant field emits many.

---

2. Composition

The six CPME mechanisms and the prior CMD + VGHT stack all operate on distinct observables and compose additively:

Layer	Mechanism	State variable
Substrate	CMD Curie gate	T_i
Substrate	VGHT sub-bands	ω_m
Substrate	VGHT voltage gate	$\|x_a\| \cdot \|x_b\|$
Substrate	VGHT viscosity	E_{total}
Injection	**CPME chirality**	$t_{\text{arrival},i}$
Filter	**CPME momentum**	E_i, E_i^{prev}
Filter	**CPME acoustic mass**	M_i
Filter	**CPME settle**	L
Metric	**CPME coherence rise**	$R_{\text{local}}, \bar{R}$
Emission	**CPME phase collapse**	$\phi_i, \phi_{\text{carrier}}$

The tick loop retains its no-if/else contract: chirality is a recorded scalar, momentum is EMA arithmetic, acoustic mass is Ornstein–Uhlenbeck, settle coupling is a sigmoid, coherence is a complex-sum norm, and phase collapse is a Gaussian window over a swept center. All are pure scalar or complex-scalar operations.

---

3. Emission Contract

At the end of each turn (i.e., after propagation ticks have run), the substrate exposes a single function:

`emit()` → `ordered list of cavity ids`

whose implementation is the Phase-Collapse sweep of §1.6. There is no `readout_slot_winners()` remaining, no S/V/O slot layer, no `syntax_slots_enabled` flag. Word order is arg-max under the Gaussian sweep window; slot count is not a parameter.

Backwards compatibility is preserved as a thin adapter: callers that previously expected a three-element slot list receive the first three emissions of the sweep, padded with `-1` if fewer are produced.

---

4. Parameters

Symbol	Default	Meaning
β	0.9	Momentum EMA smoothing factor
γ	0.30	Acoustic mass accumulation rate
δ	0.08	Acoustic mass decay rate
M_{scale}	1.0	Acoustic mask half-point
L_{thresh}	0.5	Settle-separation threshold
L_{scale}	0.2	Settle-separation sharpness

Symbol	Default	Meaning
λ	0.98	Coherence EMA baseline decay
K_{steps}	12	Phase-collapse sweep resolution
κ_{sweep}	4.0	Gaussian window sharpness
θ_{emit}	0.05	Emission floor as fraction of peak w

Ten parameters, all continuous, all with default values that recover CMD-only behaviour when set to their extreme values (e.g., $K_{\text{steps}} = 3$, $\gamma = 0$, $\kappa_{\text{sweep}} \to \infty$ reduce CPME to the three-slot readout).

---

5. Expected Dynamics

Prompt. `metformin activates receptor.`

1. **Injection** ($t = 0$ through $t_{\text{arrival, receptor}} + \Delta$): `metformin`, `activates`, `receptor` cavities are driven in that temporal order. Their t_{arrival} values differ by the inter-byte interval, encoding the chirality gradient. Acoustic mass M grows on each; their emission masks μ approach zero. Injection load L is high, so $K_{\text{settle}} \approx 0$ — the emission readout is silent regardless of field state.
2. **Propagation** (~10-40 ticks): Drive envelopes decay. Energy propagates through the graph's `REL_ACTIVATES` tunnels and VGHT voltage-gated bridges. Downstream cavities (`ampk`, `regulates`, `metabolism`, `pancreas`, etc., depending on ontology) rise in amplitude. Because they are absorbing energy from propagation rather than external drive, their momentum $P > 0$ and their acoustic mass $M \approx 0$.
3. **Settle** (~40-80 ticks): Drive envelopes decay to near zero. L falls below threshold, $K_{\text{settle}} \to 1$. Localized coherence R_{local} has risen well above $\bar{R}$; $\Delta R > 0$ signals structural resonance.
4. **Emission** (phase-collapse sweep): The carrier phase sweeps $[0, 2\pi)$. Directly-driven cavities have $\mu \approx 0$ and are excluded from arg-max even if their phase would align. Downstream cavities with $P > 0$ and $\mu \approx 1$ compete on $\phi_i \approx \phi_{\text{carrier}}$; the ordering of their phases (from their integrated positions and velocities) becomes the emission order. Output: three to four tokens describing the *consequence* of the query, not the query itself.

---

6. Failure Modes

Five known modes, all inherited from the parent proposals:

1. **Sparse-graph silence.** If no cavity outside the driven set develops positive momentum before settle, the sweep produces no emissions above floor. Output is silence, which is the correct honest response — the substrate has nothing consequential to contribute. Users habituated to always-something outputs will initially misread as regression.
2. **Chirality–Hebbian mismatch (HTR §5.2).** Chirality on injection is transient unless memory consolidation is also phase-aware. The tunnel Hebbian rule remains velocity-based; a phase-coincidence term $\Delta w_{ij} \propto v_i v_j \cos(\theta_i - \theta_j)$ would close the loop and is proposed as future work (CPME-II).

3. **Acoustic mass leakage.** If δ is too small, driven cavities stay muted across turns, and the substrate cannot reprise a recently-mentioned concept even when the new prompt properly warrants it. If δ is too large, the muting window is too short and the parrot effect returns. Default $\delta = 0.08$ places the mute window at ~ 12 ticks, comfortably shorter than a turn-to-turn interval but longer than a propagation window.
4. **Coherence baseline gaming.** $\bar{R}$ is EMA-tracked, so a long streak of high-R turns raises the baseline and reduces ΔR on subsequent similar queries. This is thermodynamically correct — habituation to the familiar — but makes ΔR absolute values non-portable across sessions. ΔR should only be compared within a session.
5. **Sweep discretization loss.** $K_{\text{steps}} = 12$ gives $\pi/6 \approx 30°$ phase resolution per step. Fields with many cavities of similar ϕ may emit fewer distinct tokens than they otherwise could. Raising K_{steps} to 24 halves the loss at doubled sweep cost.

---

7. Biological Analog

CPME's six mechanisms have direct analogs in cortical circuit architecture:

- **Chirality** corresponds to input phase-of-firing coding, well documented in hippocampal theta-cycle sequencing.
- **Momentum filter** corresponds to the divisive-normalization post-input suppression seen in early sensory areas.
- **Acoustic mass** corresponds to short-term adaptation of firing rate in cortical pyramidal cells, mediated by after-hyperpolarizing currents whose time constants match CPME's default M_i decay.
- **Settle separation** corresponds to the alpha-gamma phase separation between input reception and output articulation.
- **Coherence-rise metric** corresponds to phase-consistency measures used in cognitive neuroscience as correlates of stimulus binding.
- **Phase-collapse sweep** corresponds to phase-of-firing readout from downstream cortical areas.

CPME does not claim to model any of these processes. It claims that a substrate whose emission layer exhibits the *dynamical signatures* of all six simultaneously is the minimum architecture on which non-parroting, variable-length, phase-ordered generation is stable.

---

8. Contract Summary

The tick loop remains free of if/else on relations, domain tags, cluster ids, or any string-valued metadata. The emission function is one Gaussian window over a swept center under three multiplicative eligibility filters (momentum, acoustic mass, settle coupling), with an emission floor. All ten CPME parameters are continuous scalars. Default values recover CMD three-slot readout in the appropriate limit. All defaults *ON*.

CPME is not a feature flag. It is the emission substrate of the CMD-VGHT-CPME engine.

---

9. Summary

CPME merges MCPE's momentum-and-settle filtering with HTR's chiral injection and phase-collapse projection into a single emission architecture on a shared phase manifold. Word order is

encoded on injection as arrival-phase chirality; structural resonance is measured by localized coherence rise; eligibility is filtered by momentum, acoustic mass, and settle coupling; emission is a Gaussian-windowed sweep of a carrier phase across the per-cavity phase field. The result is a substrate whose emission is simultaneously non-parroting (no cavity that just spiked can win a sweep step), variable-length (as many tokens as the field has distinct phase peaks), word-ordered by physics (per-cavity $\phi_i = \text{atan2}(v_i, \omega_0 x_i)$), and honestly silent when its graph is not structurally resonant with the input. No syntax trees, no next-token statistics, no fixed slots.